JM0211 Process Mining

Conformance Checking for Process Improvement

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Motivation and Scientific Background

- **Use case.** Dutch bank, BPI 2017: 3,093 denials, under 1 day to ~260 days
- **Question.** Which deviations go with slow denials? Allow or forbid?
- **Not all bad.** Some are workarounds (Dees et al., 2017)
- **Course fit.** We check **control flow vs a model**

3,093	54	14.43	fast 1,501 / slow 1,592
denied applications	visible activities	days, median throughput	cutoff 14 days

Explanation of Conceptual Approach

Input **XES log** + **Petri net** (PNML) → **Output** **Rules** + **trace lists** (CSV)

1 KPI Throughput time → **2 Align, encode** Count log/model moves → **3 Classify** Slow > 14 days; tree → **4 Rules** Leaf = rule

Step 2: trace A, A, C vs model A, B, C

	1	2	3	4
Trace	A	A	>>	C
Model	A	>>	B	C
Move	sync	**log move A**	**model move B**	sync

`log_move_A = 1, model_move_B = 1 → 54 x 2 = 108 features`

Parameters

Cutoff	14 days
Hold-out	25%, seed 0
Grid	48 combinations
Tuning	5-fold CV, F1 of slow

Evaluation Design and Results

1. Correctness: hand-built gold standard

2 toy logs (5 and 6 traces), derived by hand

5/5, 6/6 traces exact | **1.0** alignment P, R | **1.0** rule P, R

2. Usefulness: real log, 25% hold-out

Slow class vs **all-slow baseline**, seeds 0 to 3

0.712 F1 seed 0 | **0.688** mean | **0.680** baseline

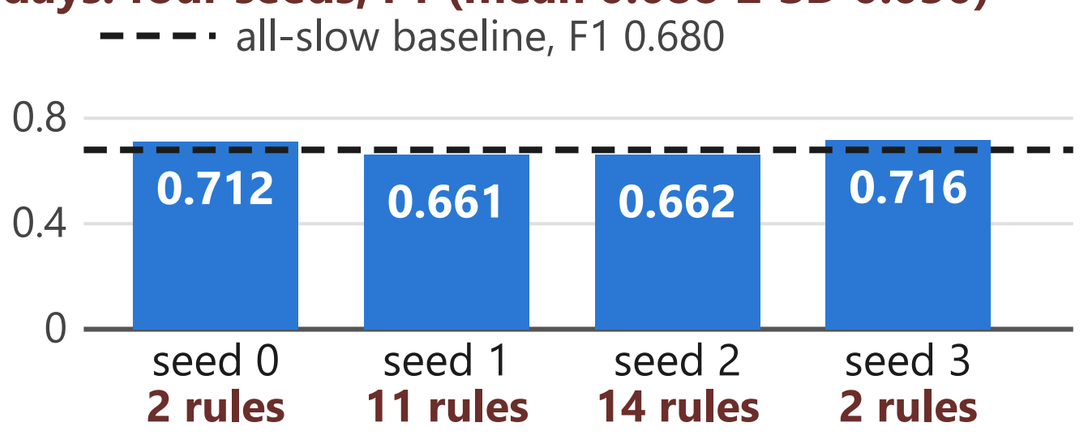
Seed 0 tree: two rules, one split

log move of `W-Call-after-offers-suspend` ?

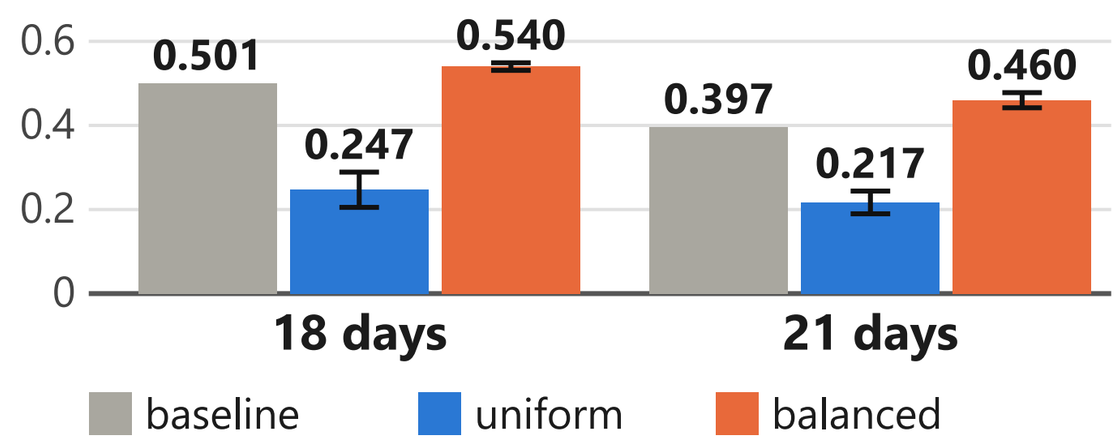
none	at least one
389 traces, 16% slow **fast**	**2,704** traces, 57% slow **slow**

Base rate of slow: 51.5%. P 0.563, R 0.970, F1 0.712

14 days: four seeds, F1 (mean 0.688 ± SD 0.030)



18 and 21 days: class weights (F1)



Correct on the gold standard, but the deviation-to-delay link is weak and unstable.

Limitations and Future Work

Limitations

- Tree unstable across splits
- Root split hits 87% of traces: net too strict?
- One cutoff, one dataset, control flow only

Future work

- Regression tree on throughput time
- Several optimal alignments per trace
- Data-aware checking (e.g. loan amount)

References: Dees, de Leoni, Mannhardt (2017). van der Aalst (2016) Process Mining. Berti et al. (2019) PM4Py. van Dongen (2017).